NURBEK TASTAN

nurbek.tastan@mbzuai.ac.ae ⋄ +971 (54) 346-9180
website: tnurbek.github.io ⋄ linkedin: nurbek-tastan ⋄ github: tnurbek

PROFILE

Doctoral researcher in machine learning with a strong foundation in probabilistic modeling, optimization, and scalable systems. My research focuses on **trustworthy federated learning**, with particular emphasis on **privacy, fairness, efficiency**, and **robustness** in collaborative settings. I also develop methods to make large-scale models, including **language models**, more efficiently fine-tunable while also preserving utility and user confidentiality.

Passionate about building machine learning systems that are not only performant, but also **privacy-aware**, **equitable**, and **computationally viable** for real-world deployment.

EDUCATION

PhD in Machine Learning, Mohamed bin Zayed University of Artificial Intelligence — 2023 - Present
Abu-Dhabi, United Arab Emirates
Relevant Coursework: Foundations and Advanced Topics in Machine Learning, Advanced Probabilistic and Statistical Inference, Advanced Topics in Continuous Optimization, Federated Learning, Safe and Robust Computer Vision, Advanced Machine Learning

MSc in Machine Learning, Mohamed bin Zayed University of Artificial Intelligence — 2021 - 2023
Abu-Dhabi, United Arab Emirates
Relevant Coursework: Mathematical Foundations for AI, Machine Learning, Advanced ML, Trustworthy AI, Probabilistic and Statistical Inference, Causality, Reinforcement Learning, Optimization

BSc in Systems of Information Security, International Information Technology University — 2017 - 2021
Almaty, Kazakhstan
Relevant Coursework: Programming (Python, C++, Java), ML, Cybersecurity, a lot of Maths
GPA **3.94**/4.00 (Top 1st of 2021 graduating class (out of 900 students))

EXPERIENCE

Lead Teaching Assistant — Jan 2022 - Present
MBZUAI — *Abu-Dhabi, United Arab Emirates*

- Courses: [DS702: Big Data Processing], [MTH701: Mathematical Foundations of Artificial Intelligence]
- Led sessions for graduate courses, covering distributed data systems and mathematical foundations for AI.
- Designed problem sets and lab materials in probability theory, statistical inference, and scalable ML systems.
- Supervised student projects and delivered guidance on Spark, Hadoop, PageRank, LSH, and big data workflows.
- Enhanced student understanding through hands-on tutorials and assessment-driven feedback.
- More information on https://tnurbek.github.io/teaching/.

Data Scientist — May 2020 - Sep 2021
InCyberService (Healthcare) — *Almaty, Kazakhstan*

- Developed classification models for highly imbalanced healthcare data.
- Built OCR-based object detection pipelines for extracting clinical information from scanned documents.
- Designed and maintained a big data pipeline using Apache Spark, Airflow, and Hadoop.

Python Developer and Teacher — Jan 2018 - Mar 2019
Bolashak School, IITU — *Almaty, Kazakhstan*

- Built a full-stack school management system using Python and Django to streamline internal workflows.
- Taught programming fundamentals and advanced topics (OOP, web scraping, algorithms) in Python and C++.

SKILLS

Languages	Python, C++, Scala, SQL (PostgreSQL, ClickHouse, Cassandra)
Frameworks	PyTorch, Tensorflow, Spark/PySpark, HuggingFace Transformers
Research Focus	Federated Learning, Decentralized Learning, Trustworthy Machine Learning, Privacy, Fairness, Efficient Fine-tuning of Large Models, Optimization, Distributed Optimization, Efficient Machine Learning, Sparsity in LLMs, Data Valuation, Differential Privacy
Additional Tools	**Big Data:** Apache Spark, PySpark, Hadoop, Kafka, Airflow, Spark Streaming, **Tools & Infrastructure:** Docker, Git, Unix/Linux, REST API, TensorBoard, W&B

PUBLICATIONS

LoFT: Low-Rank Adaptation That Behaves Like Full Fine-Tuning.
Nurbek Tastan, *Stefanos Laskaridis, Martin Takáč, Karthik Nandakumar, Samuel Horváth.*
NeurIPS, 2025. *Under Review.*

Aequa: Provably Fair Federated Learning using Slimmable Networks.
Nurbek Tastan, *Samuel Horváth, Karthik Nandakumar.* ICML, 2025.

A Framework for Double-Blind Federated Adaptation of Foundation Models.
Nurbek Tastan, *Karthik Nandakumar.*
ICLR 2025 Workshop on Modular, Collaborative, and Decentralized Deep Learning (MCDC@ICLR25).
ICCV, 2025. *Under Review.*

CYCle: Choosing Your Collaborators Wisely to Enhance Collaborative Fairness in Decentralized Learning.
Nurbek Tastan, *Samuel Horváth, Karthik Nandakumar.* TMLR, 2025. *Under Review.*

FedPeWS: Personalized Warmup via Subnetworks for Enhanced Heterogeneous Federated Learning.
Nurbek Tastan, *Samuel Horváth, Martin Takáč, Karthik Nandakumar.* CPAL, 2025.

Redefining Contributions: Shapley-Driven Federated Learning.
Nurbek Tastan, *Samar Fares, Toluwani Aremu, Samuel Horváth, Karthik Nandakumar.* IJCAI, 2024.

Collaborative Learning of Anomalies with Privacy (CLAP) for Unsupervised Video Anomaly Detection: A New Baseline.
Anas Al-lahham, Muhammad Zaigham Zaheer, ***Nurbek Tastan***, *Karthik Nandakumar.* CVPR, 2024.

A Coarse-to-Fine Pseudo-Labeling (C2FPL) Framework for Unsupervised Video Anomaly Detection.
Anas Al-lahham, ***Nurbek Tastan***, *Muhammad Zaigham Zaheer, Karthik Nandakumar.* WACV, 2024.

CaPriDe Learning: Confidential and Private Decentralized Learning based on Encryption-friendly Distillation Loss.
Nurbek Tastan, *Karthik Nandakumar.* CVPR, 2023.

Valid and Invalid Bitcoin Transactions.
Saule Amanzholova, ***Nurbek Tastan***, *Kamila Kalkamanova, Amina Yessenalina.* ACM ICEMIS, 2020.

Burglary Detection Framework for House Crime Control.
Nurbek Tastan, *Abdul Razaque, Mohamed Frej, Saule Amanzholova, R. Ganda, F. Amsaad.* IEEE ICCSA, 2019.

HONORS & AWARDS

National & Prestigious Awards

Presidential Scholarship, Ministry of Education, Kazakhstan	2020
Award "Altyn Belgi" (Golden Badge), Ministry of Education, Kazakhstan	2017
National Presidential Olympiad, Ministry of Education, Kazakhstan (I place - regional, III place - final)	2016
Republican Scientific Competition (Applied Mathematics) - III place (Bronze)	2016
National Scientific Competition (Computer Science) - II place (Silver)	2020, 2021
National Mathematics competition among students, Kazakhstan, IV place	2019

International & Technical Competitions

ICPC regional, II place — 2019

"Hack The Space" Hackathon, Dubai, United Arab Emirates, "The Best Math Team" — 2022

University- or Region-level Competitions

Scientific competition (Computer Science), IITU, Kazakhstan, II place — 2021

Scientific competition (Computer Science), IITU, Kazakhstan, III place — 2020

Mathematics competition, IITU, I place — 2019

Mathematics competition, IITU, II place — 2018

Physics Olympiad, II place (Silver) — 2015

Regional scientific competition (Applied Mathematics), I place — 2015

Mathematics Olympiad, I place (Gold) — 2013

CONFERENCE / WORKSHOP PRESENTATIONS

– Aequa: Provably Fair Federated Learning using Slimmable Networks. **ICML**, Vancouver, Canada, Jul 2025.

– A Framework for Double-Blind Federated Adaptation of Foundation Models. **ICLR MCDC Workshop**, Singapore, Apr 2025.

– FedPeWS: Personalized Warmup via Subnetworks for Enhanced Heterogeneous Federated Learning. **CPAL**, Stanford, USA, Mar 2025.

– Confidential, Private, and Fair Decentralized Learning. **ADIA Symposium**, Abu Dhabi, UAE, Nov 2024.

– Redefining Contributions: Shapley-Driven Federated Learning. **IJCAI**, Jeju, South Korea, Aug 2024.

– Collaborative Learning of Anomalies with Privacy (CLAP) for Unsupervised Video Anomaly Detection: A New Baseline. **CVPR**, Seattle WA, USA, Jun 2024.

– A Coarse-to-Fine Pseudo-Labeling (C2FPL) Framework for Unsupervised Video Anomaly Detection. **WACV**, Waikoloa, Hawaii, USA, Jan 2024.

– CaPriDe Learning: Confidential and Private Decentralized Learning based on Encryption-friendly Distillation Loss. **2nd MBZUAI CL Workshop**, Abu Dhabi, UAE, Dec 2023.
CVPR, Vancouver, Canada, Jun 2023.

ACADEMIC SERVICES

Program Committee (PC) / Reviewer for:

Conference: AAAI 2024, ICLR 2025, CVPR 2025, ICML 2025, ICCV 2025, NeurIPS 2025

Journal: IEEE Transactions on Mobile Computing (TMC)

CERTIFICATES

Machine Learning Specialization by DeepLearning.AI — Coursera, Dec, 2022

- Supervised Machine Learning: Regression and Classification
- Advanced Learning Algorithms
- Unsupervised Learning, Recommenders, Reinforcement Learning

Cybersecurity by IBM — Coursera, Jun, 2020

REFERENCES

Dr. Karthik Nandakumar
Primary Supervisor
Associate Professor at MBZUAI & MSU
karthik.nandakumar@mbzuai.ac.ae, nandakum@msu.edu

Dr. Samuel Horváth
Secondary Supervisor
Assistant Professor at MBZUAI
samuel.horvath@mbzuai.ac.ae